Abstract Algebra - Homework 2

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Problem 1

(1)
$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence, $ij \neq ji$, so multiplication is not commutative and $\mathbb{H}$ is a non-commutative ring.

(2) **(a)** Let $x, y \in \mathbb{H}$. One can verify by direct computation that conjugation reverses the order of multiplication, $\overline{xy} = \bar{y}\bar{x}$. Note that for any $h = a + bi + cj + dk \in \mathbb{H}$, we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so $N(h)$ lies in the center of $\mathbb{H}$ and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

(b) Since $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$, it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus, $N(x) = 0$ if and only if $x = 0$.

(3) Let $x \in \mathbb{H}$ be nonzero. From previous section, we have $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2 > 0$. Define the inverse of x by

$$x^{-1} := \frac{\bar{x}}{N(x)}.$$

Then:

$$xx^{-1} = x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1, \quad x^{-1}x = \frac{\bar{x}}{N(x)} \cdot x = \frac{\bar{x}x}{N(x)} = \frac{N(x)}{N(x)} = 1.$$

Hence, every nonzero element of $\mathbb{H}$ is invertible.

Suppose $x, y \in \mathbb{H}$ with $x \neq 0$ and $xy = 0$. Then, $x^{-1}(xy) = (x^{-1}x)y = 1 \cdot y = y = 0$. Thus, $y = 0$, and similarly one shows that $x = 0$ if $y \neq 0$ and $xy = 0$. Therefore, $\mathbb{H}$ has no zero-divisors, and every nonzero element is invertible.

(4) *Note: The problem appears to contain a typo, We interpret* $H = \mathbb{Z} \cdot i + \mathbb{Z} \cdot j + \mathbb{Z} \cdot k + \mathbb{Z} \cdot w$.

Since each generator lies in $\mathbb{H}$, we have $H \subset \mathbb{H}$. The set H is closed under addition and subtraction because it is a $\mathbb{Z}$-module. Moreover,

$$1 = 2w - (i + j + k) \in H,$$

so H contains the identity and hence also 0.

To verify closure under multiplication, note that the products of i, j, k lie in the $\mathbb{Z}$-span of i, j, k, and known identities in the Hurwitz quaternions show that products involving w, such as w^2, iw, jw, and kw, also lie in the $\mathbb{Z}$-span of i, j, k, w. This can be verified by direct computation.

Therefore, H is a subring of $\mathbb{H}$. Since $\mathbb{H}$ has no zero-divisors and $H \subset \mathbb{H}$, it follows that H also has no zero-divisors.

(5) Fix $a, b \in H$, with $b \neq 0$. We aim to show that there exist $q, r \in H$ such that

$$a = qb + r \quad \text{with } r = 0 \text{ or } N(r) < N(b),$$

and similarly on the right: $a = bq + r$ with the same condition on r.

Since $\mathbb{H}$ is a division ring, we can write $a = q'b + r'$ for some $q' \in \mathbb{H}$, where $r' \in \mathbb{H}$. Let $q \in H$ be the Hurwitz quaternion closest to q' in the standard Euclidean norm on $\mathbb{R}^4$. Define

$$r := a - qb \in \mathbb{H}.$$

Then $N(r) = \|a - qb\|^2$ is minimized by construction, and since H is closed under subtraction, $r \in H$. Moreover, unless $q = q'$, we have $N(r) < N(b)$ because the minimal distance approximation ensures the remainder is strictly shorter than b in norm.

The same argument applies for right division by expressing $a = bq'$, choosing the nearest $q \in H$, and setting $r := a - bq$. Thus, H is both left and right Euclidean with respect to the norm N.

Problem 2

(1) We factor the polynomial:

$$f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3),$$

which has roots $\pm\sqrt{2}, \pm\sqrt{3}$. These roots are not in $\mathbb{Q}$, so the splitting field is

$$E = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

We compute the degree using the tower law:

$$\begin{aligned}
&[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = \\
&[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \\
&2 \cdot 2 = 4.
\end{aligned}$$

(2) Let $\alpha = \sqrt{2} + \sqrt{3}$. Then

$$\alpha^2 = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6},$$

so

$$\sqrt{6} = \frac{\alpha^2 - 5}{2} \in \mathbb{Q}(\alpha).$$

Using $\sqrt{6} = \sqrt{2} \cdot \sqrt{3}$, we can write

$$\sqrt{2} = \frac{\sqrt{6}}{\sqrt{3}}, \quad \text{and} \quad \sqrt{3} = \alpha - \sqrt{2}.$$

Thus $\sqrt{2}, \sqrt{3} \in \mathbb{Q}(\alpha)$, so

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}(\alpha).$$

On the other hand, $\alpha = \sqrt{2}+\sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$ (since the field contains both $\sqrt{2}$ and $\sqrt{3}$, and is closed under addition), so the reverse inclusion holds. Therefore,

$$\mathbb{Q}(\sqrt{2}+\sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

Problem 3

The polynomial $X^2 + 1$ is irreducible over $\mathbb{F}_3$, so the quotient ring $K = \mathbb{F}_3[X]/(X^2 + 1)$ is a field with $3^2 = 9$ elements. Each element of K has a unique representative of the form $a + bX$, with $a, b \in \mathbb{F}_3$. Since $X^2 \equiv -1 \equiv 2$ in $\mathbb{F}_3$, it follows that $X^2 = 2$ in K.

The multiplicative group $K^* = K \setminus \{0\}$ has order 8 and is cyclic. To identify a generator, consider the element $X + 1 \in K$. The successive powers of $X + 1$ modulo $X^2 + 1$ are computed as follows, using the relation $X^2 = 2$:

$$(X + 1)^1 = X + 1$$
$$(X + 1)^2 = X^2 + 2X + 1 = 2 + 2X + 1 = 2X + 3 \equiv 2X$$
$$(X + 1)^3 = 2X(X + 1) = 2X^2 + 2X = 4 + 2X \equiv 1 + 2X$$
$$(X + 1)^4 = (1 + 2X)(X + 1) = X + 1 + 2X^2 + 2X = X + 1 + 4 + 2X = 3X + 5 \equiv 2$$
$$(X + 1)^5 = 2(X + 1) = 2X + 2$$
$$(X + 1)^6 = (2X + 2)(X + 1) = 2X^2 + 2X + 2X + 2 = 4 + 4X \equiv 1 + X$$
$$(X + 1)^7 = (1 + X)(X + 1) = X + 1 + X^2 + X = X + 1 + 2 + X = 2X + 3 \equiv 2X$$
$$(X + 1)^8 = ((X + 1)^4)^2 = 2^2 = 4 \equiv 1$$

The element $X + 1$ satisfies $(X + 1)^8 = 1$, and no smaller positive power yields 1. Therefore, $X + 1$ has order 8 in K^*, and is a generator.

Problem 4

(1) For all $x \in A$,

$$(x + x)^2 = x + x,$$

since $x + x \in A$ and the ring is Boolean. Expanding the left-hand side gives:

$$(x + x)^2 = x^2 + x^2 + x^2 + x^2 = 4x.$$

Thus:

$$4x = x + x \Rightarrow 3x = 0 \Rightarrow 2x = 0.$$

Hence, $2x = 0$ for all $x \in A$.

(2) Let $\mathfrak{p} \subset A$ be a prime ideal. Then the quotient ring $A/\mathfrak{p}$ is an integral domain. Moreover, since the property $x^2 = x$ is preserved under quotienting, $A/\mathfrak{p}$ is also Boolean.

In a Boolean ring, every element satisfies $x^2 = x$, so $x(x-1) = 0$. In an integral domain, this implies $x = 0$ or $x = 1$, since there are no zero-divisors. Therefore, $A/\mathfrak{p}$ contains only two elements: 0 and 1, and hence is isomorphic to the field $\mathbb{F}_2$.

It follows that $\mathfrak{p}$ is maximal, since the quotient $A/\mathfrak{p}$ is a field.

(3) Let $I = (x_1, x_2, \dots, x_n)$ be a finitely generated ideal in A. Define:

$$e := x_1 + x_2 + \cdots + x_n.$$

Since A is Boolean, $e^2 = e$, so $e \in A$. We claim that $I = (e)$.

First, each $x_i \in I$, and since $x_i^2 = x_i$, we have $x_i = x_i e \in (e)$, so $x_1, \dots, x_n \in (e)$, hence $I \subseteq (e)$.

Conversely, $e = x_1 + \cdots + x_n \in I$, so any multiple $ae \in I$, and thus $(e) \subseteq I$. Therefore, $I = (e)$, and the ideal is principal.

(4) Yes, there exists a Boolean ring which is not a principal ideal domain. For example, consider the Boolean ring $A = \mathcal{P}(\mathbb{N})$, the power set of the natural numbers, with addition defined as symmetric difference and multiplication as intersection. This is a Boolean ring, since every subset satisfies $S^2 = S \cap S = S$.

Define the ideal $I \subset A$ generated by all singleton subsets:

$$I = \langle \{n\} \mid n \in \mathbb{N} \rangle.$$

This ideal is not principal, as no single subset can generate all singleton sets under symmetric difference and intersection. Therefore, A is a Boolean ring that is not a PID.

Problem 5

(1) Let A be a UFD and let $S, T \in A[X]$ be primitive polynomials. Suppose for contradiction that ST is not primitive. Then there exists an irreducible element $p \in A$ such that p divides every coefficient of ST.

Consider the canonical ring homomorphism $\varphi : A \to A/(p)$, and extend it to $\varphi : A[X] \to (A/(p))[X]$ by applying it to each coefficient. Since p is irreducible in a UFD, it generates a prime ideal, so $A/(p)$ is an integral domain, and thus $(A/(p))[X]$ is also an integral domain.

Then $\varphi(ST) = \varphi(S)\varphi(T) = 0$ in $(A/(p))[X]$. But since S and T are primitive, not all their coefficients are divisible by p, so $\varphi(S) \neq 0$ and $\varphi(T) \neq 0$. This contradicts the fact that the product of two nonzero elements in a domain cannot be zero.

Therefore, ST must be primitive.

(2) Let A be a UFD. By previous section, the product of two primitive polynomials in $A[X]$ is primitive. Together with the fact that every nonzero polynomial $F \in A[X]$ can be written uniquely (up to units) as $F = c \cdot F_0$, where $c \in A$ and F_0 is primitive, it follows that irreducibility and factorization in $A[X]$ reduce to those in A and in $A[X]$ over its field of fractions.

Since A is a UFD and its field of fractions $\mathrm{Frac}(A)$ makes $\mathrm{Frac}(A)[X]$ a UFD, it follows that $A[X]$ is also a UFD.

Problem 6

Since $h(X) \in I$ is nonconstant with constant term 1, write

$$h(X) = a_0 + a_1 X + \cdots + a_n X^n, \quad \text{with } a_0 = 1,\ a_i \in \mathbb{Z}.$$

Then define $f(X) := a_1 + a_2 X + \cdots + a_n X^{n-1} \in \mathbb{Z}[X]$, so that

$$h(X) = 1 + X f(X),$$

with $f(X) \in \mathbb{Z}[X]$.

We claim that for each $r \in \mathbb{N}$, there exist polynomials $a_r(X), b_r(X) \in \mathbb{Z}[X]$ such that:

$$S_r = u(X) a_r(X) + X^{r+1} b_r(X).$$

This follows from the fact that $u(X) \equiv 1 \mod X$, so it is invertible modulo X^{r+1}, and therefore:

$$S_r \equiv u(X)a_r(X) \mod X^{r+1} \quad \text{for some } a_r(X),$$

yielding the desired decomposition for some $b_r(X) \in \mathbb{Z}[X]$.

Now, since $u(X) \in I$, to conclude that $S_r \in I$, it suffices to show that $X^{r+1}b_r(X) \in I$ for some $r \in \mathbb{N}$. To that end, note that since the elements of I have no nonconstant common divisor, the ideal $I \cap \mathbb{Z} \subset \mathbb{Z}$ must be nontrivial. That is, I contains a nonzero integer $\delta \in \mathbb{Z} \setminus \{0\}$.

Because $b_r(X) \in \mathbb{Z}[X]$, the set $\{b_r(X) \mod \delta\}$ is finite modulo δ, so there exist integers $r < s$ such that:

$$b_s(X) \equiv b_r(X) \mod \delta \quad \Rightarrow \quad b_s(X) - b_r(X) \in (\delta) \subset I.$$

It follows that:

$$S_s - S_r = X^{r+1}(b_s(X) - b_r(X)) \in I,$$

and since $S_r \in I$ up to the correction term $X^{r+1}b_r(X)$, we conclude $S_s \in I$.

Therefore, for some $r \in \mathbb{N}$, the polynomial $1 + X + X^2 + \cdots + X^r \in I$, as required.